# LESSON PLAN

## LSF, STABILITY, A/C PERFORMANCE (6)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 06 |
| **Total No. of Lesson Plans** | 33 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Boundary layer, laminar, transition, turbulent B/L |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Boundary layer, laminar, transition, turbulent B/L, |
| **5. Specific Objectives** | 6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of IB diagrams showing flow separation and reattachment points; Instructor supervision required for diagram interpretation. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What is the definition of a boundary layer?
(b) How does viscosity affect flow over a body?
(c) Why is the boundary layer important for aircraft performance?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition of boundary layer<br>• No-slip condition at surface<br>• Velocity gradient profile<br>• Thickness of boundary layer<br>• Displacement thickness concept | L & Ds | PPT, IB | 15 min |
| (b) | • Laminar flow characteristics<br>• Turbulent flow characteristics<br>• Transition region definition<br>• Reynolds number influence<br>• Skin friction comparison | L & Ds | PPT, PDF | 20 min |
| (c) | • Effects of boundary layer on drag<br>• Flow separation mechanisms<br>• Impact on aircraft efficiency<br>• Control surface effectiveness<br>• Stability implications | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between laminar and turbulent flow.
(b) Describe how the boundary layer affects skin friction drag.
(c) What factors influence the transition from laminar to turbulent flow?

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| **12. Home Assignment and Follow Up** | Prepare a brief note on how boundary layer control techniques can improve aircraft performance, referencing concepts from LP 02. |

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| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Geometrical characteristics of aerofoil, types of aerofoil, definition of terms. | Refer to the master syllabus reference list | LP 07 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min